Paper 3: R console output

Migration Concentration and Financial Network Embeddedness, moderated by Ethnic Diversity and Creative-Class Share (Australian SA4 regions)

This is the unedited R console transcript of the Paper 3 Lagged (primary specification) pipeline run, as saved from the author's R session (R 4.5.3, Windows). It includes the echoed code, model output, diagnostics, correlation tables and the N = 80 raw descriptive statistics. Nothing has been altered, re-run or reformatted other than line wrapping for the page width.

Source file: console log of 26 Oct 2026. Contains local file paths from the author's computer.

```
R version 4.5.3 (2026-03-11 ucrt) -- "Reassured Reassurer"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

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  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> # ============================================================================
> # PAPER 3 - LAGGED PIPELINE (Option D: primary specification, NATIVE PAIRING)
> # This version's Model 4 uses each table's own theorized IV only (SE1 alone
> # in Table 3A, SE2 alone in Table 3B) - matching the final dissertation
> # tables. Two path fixes applied: migration files (.xlsx, new location) and
> # All SA4s.data.csv (moved to Desktop).
> # ============================================================================
> library(tidyverse)
── Attaching core tidyverse packages ──────────────
```

```
> library(lmtest)
Loading required package: zoo

Attaching package: ‘zoo’

The following objects are masked from ‘package:base’:

    as.Date, as.Date.numeric
> library(sandwich)
> library(car)
Loading required package: carData

Attaching package: ‘car’

The following object is masked from ‘package:dplyr’:

    recode

The following object is masked from ‘package:purrr’:
    some
>
> path_base    <- "C:/Users/s222384959/OneDrive - Deakin University"
> path_desktop <- file.path(path_base, "Desktop")
> path_paper3  <- file.path(path_desktop, "Paper 3")
> path_docs_ee <- file.path(path_base, "Documents/EE/Au reports")
> path_teach   <- file.path(path_desktop, "Teaching")
>
> f_postal_sa4        <- file.path(path_docs_ee, "Postal to SA4.xlsx")
> f_overlap_poa        <- file.path(path_desktop, "overlap_poa_details.csv")
> f_sa4_lookup         <- file.path(path_base, "SA4 codes and names .xlsx")
> f_vc_deals            <- file.path(path_paper3, "Final VC Deals.xlsx")
> # <<FIXED>> migration files moved here, now .xlsx not .csv
> f_migration_17_21    <- file.path(path_teach, "data/Internal_migration_SA4_1621.xlsx")
> f_migration_22_25    <- file.path(path_teach, "data/Internal_migration_SA4_2225.xlsx")
> f_population          <- file.path(path_desktop, "resident_population.csv")
> f_business_churn      <- file.path(path_desktop, "Business_Churn_SA4_202124.xlsx")
> f_homeostasis_raw     <- file.path(path_teach, "data/Dependent Variable_Homeostasis.csv")
> f_uv_anzsic           <- "C:/Users/s222384959/OneDrive - Deakin University/UV_ANZSIC_2021.xlsx"
> f_mv1_2016            <- file.path(path_desktop, "MV1_ED_SA4_2016.csv")
> f_mv2_2016            <- file.path(path_desktop, "mv2_CC_2016.csv")
> f_specialisation_2016 <- file.path(path_desktop, "specialisation_index_2016_FINAL.csv")
> f_employed_pre2021   <- file.path(path_desktop, "Employed_SA4_201425.xlsx")
> f_income              <- file.path(path_desktop, "md_hh_inc.csv")
> f_age                  <- file.path(path_desktop, "demographic 25-64.csv")
> f_unemployment        <- file.path(path_desktop, "unemployment_finalrate.csv")
> f_org_parts <- file.path(path_teach, paste0("data/organizations_part_", 1:7, ".csv"))
> # <<FIXED>> moved to Desktop
> f_all_sa4_data <- file.path(path_desktop, "All SA4s.data.csv")
>
> clean_sa4 <- function(x) x %>% str_to_lower() %>% str_trim() %>% str_squish()
> zscore    <- function(x) as.numeric(scale(x))
> robust_summary <- function(model) coeftest(model, vcov = vcovHC(model, type = "HC1"))
> stars_from_p <- function(p) {
+   case_when(p < 0.001 ~ "***", p < 0.01 ~ "**", p < 0.05 ~ "*", p < 0.1 ~ "+", TRUE ~ "")
+ }
> compute_se_directed <- function(edges) {
+   if (nrow(edges) == 0) return(data.frame(sa4_code21 = character(0), SE = numeric(0)))
+   edge_weights <- edges %>% count(investor_sa4, investee_sa4, name = "w_ij")
+   g <- graph_from_data_frame(edge_weights, directed = TRUE)
+   adj <- as_adjacency_matrix(g, attr = "w_ij", sparse = FALSE)
+   nodes <- V(g)$name
+   se_values <- numeric(length(nodes)); names(se_values) <- nodes
+   for (i in nodes) {
+     neigh <- which(adj[i, ] > 0 | adj[, i] > 0)
+     k <- length(neigh)
+     if (k <= 1) { se_values[i] <- 0; next }
+     sub_adj <- adj[neigh, neigh]
+     v_i <- sum(sub_adj > 0); max_edges <- k * (k - 1)
+     se_values[i] <- ifelse(max_edges == 0, 0, v_i / max_edges)
+   }
+   data.frame(sa4_code21 = names(se_values), SE = se_values)
+ }
> fmt_cell <- function(raw_model, std_model, term_raw, term_z) {
+   rs <- robust_summary(raw_model)
+   if (!term_raw %in% rownames(rs)) return(c("", "", ""))
```

```
+   est   <- rs[term_raw, "Estimate"]
+   se    <- rs[term_raw, "Std. Error"]
+   p     <- rs[term_raw, "Pr(>|t|)"]
+   stars <- stars_from_p(p)
+   beta  <- if (!is.null(std_model) && !is.null(term_z) && term_z %in% names(coef(std_model))) {
+     sprintf("%.3f", coef(std_model)[term_z])
+   } else { "" }
+   c(sprintf("%.3f%s", est, stars), sprintf("(%.3f)", se), sprintf("[%s]", beta))
+ }
> print_results_table <- function(raw_models, std_models, model_labels, var_specs) {
+   for (v in seq_along(var_specs)) {
+     label    <- names(var_specs)[v]
+     term_raw <- var_specs[[v]]$raw
+     term_z   <- var_specs[[v]]$z
+     cat(sprintf("%-32s", label))
+     lines <- lapply(seq_along(raw_models), function(m) {
+       if (is.null(raw_models[[m]])) return(c("", "", ""))
+       fmt_cell(raw_models[[m]], std_models[[m]], term_raw, term_z)
+     })
+     cat(paste(sapply(lines, `[`, 1), collapse = " | "), "\n")
+     cat(sprintf("%-32s", ""))
+     cat(paste(sapply(lines, `[`, 2), collapse = " | "), "\n")
+     cat(sprintf("%-32s", ""))
+     cat(paste(sapply(lines, `[`, 3), collapse = " | "), "\n")
+   }
+   cat(strrep("-", 100), "\n")
+   cat(sprintf("%-32s", "Observations"))
+   cat(paste(sapply(raw_models, function(m) if (is.null(m)) "" else nobs(m)), collapse = " | "), "\n")
+   cat(sprintf("%-32s", "Adjusted R-squared"))
+   cat(paste(sapply(raw_models, function(m) if (is.null(m)) "" else sprintf("%.3f", summary(m)$adj.r.squared)), collapse = " | "), "\n")
+   cat(sprintf("%-32s", "AIC"))
+   cat(paste(sapply(raw_models, function(m) if (is.null(m)) "" else sprintf("%.1f", AIC(m))), collapse = " | "), "\n")
+   cat("\nModel columns:", paste(model_labels, collapse = " | "), "\n")
+   cat("Robust (HC1) SE in parentheses (); standardized coefficient in brackets [].\n")
+   cat("*** p<0.001  ** p<0.01  * p<0.05  + p<0.1\n\n")
+ }
>
> # PART 1
> sa4_lookup_clean <- read_excel(f_sa4_lookup) %>%
+   transmute(sa4_code = as.character(SA4_CODE_2021),
+             sa4_name = str_to_lower(str_trim(SA4_NAME_2021))) %>%
+   distinct() %>%
+   filter(!is.na(sa4_code), !is.na(sa4_name), sa4_name != "na",
+          !str_detect(sa4_code, "ZZZ"),
+          !str_detect(sa4_name, "migratory|offshore|no usual address"))
> postal_sa4_raw <- read_excel(f_postal_sa4) %>%
+   clean_names() %>%
+   select(poa_code21, sa4_code21, sa4_name21) %>%
+   mutate(poa_code21 = as.character(poa_code21)) %>%
+   distinct()
> overlap_override <- read_csv(f_overlap_poa, show_col_types = FALSE) %>%
+   clean_names() %>%
+   mutate(poa_code21 = str_pad(as.character(poa_code21), width = 4, pad = "0"))
> postal_sa4 <- postal_sa4_raw %>%
+   left_join(overlap_override, by = "poa_code21") %>%
+   filter(is.na(sa4_region) | sa4_name21 == sa4_region) %>%
+   select(poa_code21, sa4_code21, sa4_name21) %>%
+   distinct()
> stopifnot(!any(duplicated(postal_sa4$poa_code21)))
> all_sa4   <- postal_sa4 %>% distinct(sa4_code21, sa4_name21)
> all_nodes <- unique(as.character(all_sa4$sa4_code21))
> population_clean <- read_csv(f_population, show_col_types = FALSE) %>%
+   mutate(sa4_name = clean_sa4(region_name), year = as.integer(year),
+          population = `Estimated resident population`) %>%
+   select(sa4_name, year, population) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   select(sa4 = sa4_code, year, population)
New names:
• `SA4_NAME_2021` -> `SA4_NAME_2021...4`
• `AREA_ALBERS_SQKM` -> `AREA_ALBERS_SQKM...5`
• `SA4_NAME_2021` -> `SA4_NAME_2021...6`
• `AREA_ALBERS_SQKM` -> `AREA_ALBERS_SQKM...7`
> population_avg_leq2020 <- population_clean %>%
+   filter(year <= 2020) %>%
+   group_by(sa4) %>% summarise(population_avg_leq2020 = mean(population, na.rm = TRUE), .groups = "drop")
```

```
> cat(sprintf("[PART 1] postal_sa4: %d POAs, %d SA4s\n",
+             n_distinct(postal_sa4$poa_code21), length(all_nodes)))
[PART 1] postal_sa4: 2631 POAs, 90 SA4s
>
> # PART 2 - <<FIXED>> read_excel + type coercion
> migration_raw <- bind_rows(
+   read_excel(f_migration_17_21) %>% mutate(REGION_DEP = as.character(REGION_DEP), REGION_ARR = as.character(REGION_ARR)),
+   read_excel(f_migration_22_25) %>% mutate(REGION_DEP = as.character(REGION_DEP), REGION_ARR = as.character(REGION_ARR))
+ ) %>%
+   transmute(year = as.numeric(TIME_PERIOD), i = as.character(REGION_DEP),
+             j = as.character(REGION_ARR), weight = as.numeric(OBS_VALUE)) %>%
+   filter(!is.na(weight), weight > 0, i != j) %>%
+   filter(i %in% all_nodes, j %in% all_nodes) %>%
+   filter(year <= 2020)
There were 50 or more warnings (use warnings() to see the first 50)
> cat(sprintf("[PART 2] Migration data restricted to years: %d-%d\n",
+             min(migration_raw$year), max(migration_raw$year)))
[PART 2] Migration data restricted to years: 2017-2020
> flows <- migration_raw %>% group_by(year, i, j) %>%
+   summarise(x_ij = sum(weight, na.rm = TRUE), .groups = "drop")
> pair_key <- flows %>% mutate(lo = pmin(i, j), hi = pmax(i, j),
+                              direction = ifelse(i == lo, "lo_to_hi", "hi_to_lo"))
> bilateral <- pair_key %>%
+   select(year, lo, hi, direction, x_ij) %>%
+   pivot_wider(names_from = direction, values_from = x_ij, values_fill = 0) %>%
+   mutate(lo_to_hi = if ("lo_to_hi" %in% names(.)) lo_to_hi else 0,
+          hi_to_lo = if ("hi_to_lo" %in% names(.)) hi_to_lo else 0) %>%
+   transmute(year, i = lo, j = hi, w_ij = lo_to_hi + hi_to_lo) %>%
+   filter(!is.na(i), !is.na(j), i != j)
> se1_yearly <- bilateral %>%
+   transmute(year, ego = i, partner = j, w_ij) %>%
+   bind_rows(bilateral %>% transmute(year, ego = j, partner = i, w_ij)) %>%
+   group_by(year, ego) %>%
+   mutate(W_i = sum(w_ij, na.rm = TRUE), p_ij = ifelse(W_i > 0, w_ij / W_i, 0)) %>%
+   summarise(SE = sum(p_ij^2, na.rm = TRUE), .groups = "drop")
> se1_avg <- se1_yearly %>%
+   group_by(ego) %>%
+   summarise(se1_avg = mean(SE, na.rm = TRUE), .groups = "drop") %>%
+   transmute(sa4 = as.character(ego), se1_avg)
> cat(sprintf("[PART 2] se1_avg (2017-2020 only): %d regions\n", nrow(se1_avg)))
[PART 2] se1_avg (2017-2020 only): 88 regions
> network_size_migration <- bilateral %>%
+   transmute(year, ego = i, partner = j) %>%
+   bind_rows(bilateral %>% transmute(year, ego = j, partner = i)) %>%
+   distinct(year, ego, partner) %>%
+   group_by(year, ego) %>%
+   summarise(n_partners = n_distinct(partner), .groups = "drop") %>%
+   group_by(ego) %>%
+   summarise(network_size_migration_avg = mean(n_partners, na.rm = TRUE), .groups = "drop") %>%
+   transmute(sa4 = as.character(ego), network_size_migration = network_size_migration_avg / 100)
> cat(sprintf("[PART 2] network_size_migration (2017-2020 avg, /100): %d regions\n", nrow(network_size_migration)))
[PART 2] network_size_migration (2017-2020 avg, /100): 88 regions
>
> # PART 3
> vc_edges <- read_excel(f_vc_deals) %>%
+   clean_names() %>%
+   separate_rows(investee_company_zip_code, sep = "\\|") %>%
+   separate_rows(firm_investors_zip_code, sep = "\\|") %>%
+   mutate(investee_company_zip_code = trimws(as.character(investee_company_zip_code)),
+          firm_investors_zip_code   = trimws(as.character(firm_investors_zip_code))) %>%
+   left_join(postal_sa4, by = c("investee_company_zip_code" = "poa_code21")) %>%
+   rename(investee_sa4 = sa4_code21) %>%
+   left_join(postal_sa4, by = c("firm_investors_zip_code" = "poa_code21")) %>%
+   rename(investor_sa4 = sa4_code21) %>%
+   mutate(year = year(ymd(investment_date))) %>%
+   filter(!is.na(investor_sa4), !is.na(investee_sa4),
+          investor_sa4 != investee_sa4, !is.na(year))
New names:
• `` -> `...1`
• `` -> `...6`
> edges_year_all <- vc_edges %>% select(year, investor_sa4, investee_sa4) %>% distinct()
> edges_year_leq2020 <- edges_year_all %>% filter(year <= 2020)
> if (nrow(edges_year_leq2020) == 0) stop("[PART 3] FATAL: no VC deal data exists at or before 2020.")
> max_vc_year_lagged <- max(edges_year_leq2020$year)
> min_vc_year_lagged <- min(edges_year_leq2020$year)
```

```
> n_years_available <- max_vc_year_lagged - min_vc_year_lagged + 1
> cat(sprintf("[PART 3] VC deal data at/before 2020 spans %d-%d (%d years available).\n",
+             min_vc_year_lagged, max_vc_year_lagged, n_years_available))
[PART 3] VC deal data at/before 2020 spans 2014-2020 (7 years available).
> edges_10yr <- edges_year_leq2020 %>% filter(year >= max_vc_year_lagged - 9, year <= max_vc_year_lagged)
> se2_avg <- compute_se_directed(edges_10yr) %>%
+   transmute(sa4 = as.character(sa4_code21), se2_avg = SE)
> cat(sprintf("[PART 3] se2_avg (<=2020 window): %d regions with observed VC ties\n", nrow(se2_avg)))
[PART 3] se2_avg (<=2020 window): 49 regions with observed VC ties
>
> # PART 3b
> burt_constraint_raw <- edges_10yr %>%
+   count(investor_sa4, investee_sa4, name = "w_ij") %>%
+   graph_from_data_frame(directed = TRUE)
> burt_constraint <- tryCatch({
+   co <- constraint(burt_constraint_raw, weights = E(burt_constraint_raw)$w_ij)
+   data.frame(sa4 = as.character(V(burt_constraint_raw)$name), burt_constraint = as.numeric(co))
+ }, error = function(e) { data.frame(sa4 = character(0), burt_constraint = numeric(0)) }) %>%
+   filter(!is.na(burt_constraint))
> in_deals_pc_final <- edges_10yr %>%
+   group_by(investee_sa4) %>%
+   summarise(in_deals = n(), .groups = "drop") %>%
+   rename(sa4 = investee_sa4) %>%
+   left_join(population_avg_leq2020, by = "sa4") %>%
+   mutate(in_deals_pc = (in_deals / population_avg_leq2020) * 1000) %>%
+   select(sa4, in_deals_pc)
> cat(sprintf("[PART 3b] burt_constraint: %d regions | in_deals_pc: %d regions\n",
+             nrow(burt_constraint), nrow(in_deals_pc_final)))
[PART 3b] burt_constraint: 49 regions | in_deals_pc: 48 regions
> network_size_vc <- edges_10yr %>%
+   transmute(ego = investor_sa4, partner = investee_sa4) %>%
+   bind_rows(edges_10yr %>% transmute(ego = investee_sa4, partner = investor_sa4)) %>%
+   distinct(ego, partner) %>%
+   count(ego, name = "n_partners") %>%
+   transmute(sa4 = as.character(ego), network_size_vc = n_partners / 100)
>
> # PART 4 - Homeostasis WITH standardization fix
> churn_raw <- read_excel(f_business_churn) %>%
+   mutate(region = Region, year = as.integer(TIME_PERIOD)) %>%
+   filter(`Data Item` == "Total business entries (no.)") %>%
+   transmute(sa4_name = clean_sa4(region), year, entries = as.numeric(OBS_VALUE))
> real_entries <- churn_raw %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   transmute(sa4 = sa4_code, year, entries) %>%
+   filter(!is.na(sa4))
> ed_cross <- real_entries %>%
+   filter(year %in% 2021:2024) %>%
+   group_by(sa4) %>%
+   summarise(entries_cumulative = sum(entries, na.rm = TRUE), .groups = "drop") %>%
+   left_join(population_clean %>% filter(year == 2024) %>%
+               select(sa4, population_2024 = population), by = "sa4") %>%
+   mutate(ed_cross = (entries_cumulative / population_2024) * 1000)
> uv_data_2digit <- read_excel(f_uv_anzsic) %>%
+   clean_names() %>%
+   rename(sa4 = sa4_ur, industry = x2_digit_level_indp_industry_of_employment, count = count) %>%
+   select(sa4, industry, count) %>%
+   filter(!str_detect(industry, regex("nfd", ignore_case = TRUE))) %>%
+   mutate(industry_2d = case_when(
+     str_detect(industry, "Agriculture|Forestry|Fishing") ~ "A", str_detect(industry, "Mining") ~ "B",
+     str_detect(industry, "Manufacturing") ~ "C", str_detect(industry, "Electricity|Gas|Water|Waste") ~ "D",
+     str_detect(industry, "Construction") ~ "E", str_detect(industry, "Wholesale") ~ "F",
+     str_detect(industry, "Retail") ~ "G", str_detect(industry, "Accommodation|Food") ~ "H",
+     str_detect(industry, "Transport|Postal|Warehousing") ~ "I",
+     str_detect(industry, "Information|Media|Telecommunications") ~ "J",
+     str_detect(industry, "Financial|Insurance") ~ "K", str_detect(industry, "Rental|Hiring|Real Estate") ~ "L",
+     str_detect(industry, "Professional|Scientific|Technical") ~ "M",
+     str_detect(industry, "Administrative|Support") ~ "N", str_detect(industry, "Public Administration") ~ "O",
+     str_detect(industry, "Education") ~ "P", str_detect(industry, "Health") ~ "Q",
+     str_detect(industry, "Arts|Recreation") ~ "R", str_detect(industry, "Other Services") ~ "S",
+     TRUE ~ NA_character_
+   )) %>%
+   filter(!is.na(industry_2d))
> uv_cross <- uv_data_2digit %>%
+   group_by(sa4, industry_2d) %>% summarise(count = sum(count), .groups = "drop") %>%
+   group_by(sa4) %>% mutate(total_emp = sum(count), p_g = count / total_emp) %>%
```

```
+   summarise(UV = sum(p_g * log2(1 / pmax(p_g, 1e-12))), .groups = "drop") %>%
+   mutate(sa4_name = str_to_lower(str_trim(sa4))) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   transmute(sa4 = sa4_code, uv = UV) %>%
+   filter(!is.na(sa4)) %>%
+   mutate(uv_norm = (uv - min(uv, na.rm = TRUE)) / (max(uv, na.rm = TRUE) - min(uv, na.rm = TRUE)),
+          uv_nl = uv_norm - uv_norm^2)
> raw_homeo <- read_csv(f_homeostasis_raw, show_col_types = FALSE) %>%
+   clean_names() %>% rename(sa4 = sa4_region) %>%
+   mutate(sa4_name = str_to_lower(str_trim(sa4))) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   mutate(sa4 = sa4_code) %>% select(-sa4_name, -sa4_code)
> kibs_2023_cross <- raw_homeo %>% select(sa4, kibs_2023) %>% rename(kibs = kibs_2023) %>% filter(!is.na(sa4))
> homeostasis_cross <- ed_cross %>%
+   select(sa4, ed_cross) %>%
+   left_join(uv_cross %>% select(sa4, uv_nl), by = "sa4") %>%
+   left_join(kibs_2023_cross, by = "sa4") %>%
+   mutate(ed_log = log(1 + ed_cross), kibs_log = log(1 + kibs),
+          z_ed_log = as.numeric(scale(ed_log)),
+          z_uv_nl  = as.numeric(scale(uv_nl)),
+          z_kibs_log = as.numeric(scale(kibs_log)),
+          homeostasis = z_ed_log + z_uv_nl + z_kibs_log)
> cat(sprintf("[PART 4] homeostasis_cross (STANDARDIZED, 2021-2024): %d regions\n", nrow(homeostasis_cross)))
[PART 4] homeostasis_cross (STANDARDIZED, 2021-2024): 88 regions
>
> # PART 5
> org_all <- map_dfr(f_org_parts, read.csv)
> org_aus <- org_all %>%
+   filter(country_code == "AUS") %>%
+   mutate(founded_on = parse_date_time(founded_on, orders = c("ymd", "dmy", "mdy", "Y"), quiet = TRUE),
+          year_founded = year(founded_on),
+          postal_code_clean = str_extract(postal_code, "\\b\\d{4}\\b")) %>%
+   filter(year_founded >= 2021, year_founded <= 2024)
> org_sa4_valid <- org_aus %>%
+   left_join(postal_sa4, by = c("postal_code_clean" = "poa_code21")) %>%
+   filter(!is.na(sa4_code21)) %>%
+   group_by(uuid, year_founded, sa4_code21, sa4_name21) %>%
+   slice(1) %>% ungroup()
> new_firms_total <- org_sa4_valid %>%
+   distinct(uuid, sa4_code21) %>%
+   count(sa4_code21, name = "total_firms") %>%
+   transmute(sa4 = as.character(sa4_code21), total_firms)
> population_avg_2021_2024 <- population_clean %>%
+   filter(year >= 2021, year <= 2024) %>%
+   group_by(sa4) %>% summarise(population_avg_2021_2024 = mean(population, na.rm = TRUE), .groups = "drop")
> all_sa4_for_firms <- sa4_lookup_clean %>% distinct(sa4 = sa4_code)
> new_firms_percapita <- all_sa4_for_firms %>%
+   left_join(new_firms_total, by = "sa4") %>%
+   mutate(total_firms = replace_na(total_firms, 0)) %>%
+   left_join(population_avg_2021_2024, by = "sa4") %>%
+   mutate(new_firms_percapita = (total_firms / population_avg_2021_2024) * 1000)
> cat(sprintf("[PART 5] new_firms_percapita: %d total regions (%d with >=1 firm, %d zero-filled)\n",
+             nrow(new_firms_percapita), sum(new_firms_percapita$total_firms > 0), sum(new_firms_percapita$total_firms == 0)))
[PART 5] new_firms_percapita: 89 total regions (68 with >=1 firm, 21 zero-filled)
>
> # PART 6 - <<FIXED>> f_all_sa4_data now points to Desktop
> abs_regional_raw <- read_csv(f_all_sa4_data, show_col_types = FALSE)
> density_year_col <- intersect(c("TIME_PERIOD", "YEAR", "Year", "year"), names(abs_regional_raw))
> if (length(density_year_col) > 0) {
+   density_avg <- abs_regional_raw %>%
+     filter(REGIONTYPE == "SA4", MEASURE == "ERP_21", !is.na(OBS_VALUE),
+            .data[[density_year_col[1]]] <= 2020) %>%
+     mutate(sa4 = as.character(ASGS_2021)) %>%
+     group_by(sa4) %>%
+     summarise(population_density_avg = mean(OBS_VALUE, na.rm = TRUE), .groups = "drop")
+   cat(sprintf("[PART 6] Density restricted to <=2020 using column '%s'.\n", density_year_col[1]))
+ } else {
+   density_avg <- abs_regional_raw %>%
+     filter(REGIONTYPE == "SA4", MEASURE == "ERP_21", !is.na(OBS_VALUE)) %>%
+     mutate(sa4 = as.character(ASGS_2021)) %>%
+     group_by(sa4) %>%
+     summarise(population_density_avg = mean(OBS_VALUE, na.rm = TRUE), .groups = "drop")
+ }
[PART 6] Density restricted to <=2020 using column 'TIME_PERIOD'.
> mv1_2016 <- read_csv(f_mv1_2016, show_col_types = FALSE) %>%
```

```
+   clean_names() %>%
+   transmute(sa4_name = str_to_lower(str_trim(sa4_ur)), entropy = entropy_std)
> entropy_2016 <- mv1_2016 %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   transmute(sa4 = sa4_code, entropy)
> ccs_2016 <- read_csv(f_mv2_2016, show_col_types = FALSE) %>%
+   clean_names() %>%
+   mutate(sa4_name = str_to_lower(str_trim(sa4_ur))) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   transmute(sa4 = sa4_code, creative_class_share)
>
> # PART 7
> employment_panel <- read_excel(f_employed_pre2021) %>%
+   mutate(sa4_name = clean_sa4(`Region of usual residence (SA4)`), year = as.integer(year),
+          employed = as.numeric(Employed) * 1000) %>%
+   filter(year >= 2014, year <= 2020) %>%
+   select(sa4_name, year, employed) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   select(sa4 = sa4_code, year, employed)
> income_clean <- read_csv(f_income, show_col_types = FALSE) %>%
+   mutate(sa4_name = clean_sa4(sa4_name), year = as.integer(year)) %>%
+   filter(year <= 2020) %>%
+   select(sa4_name, year, log_med_income) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   select(sa4 = sa4_code, year, log_med_income)
> age_clean <- read_csv(f_age, show_col_types = FALSE) %>%
+   mutate(sa4_name = clean_sa4(region_name), year = as.integer(date),
+          age_25_64 = `Proportion of Estimated Resident Population that is 25-64 years old`) %>%
+   filter(year <= 2020) %>%
+   select(sa4_name, year, age_25_64) %>%
+   left_join(sa4_lookup_clean, by = "sa4_name") %>%
+   select(sa4 = sa4_code, year, age_25_64)
> unemployment_clean <- read_csv(f_unemployment, show_col_types = FALSE) %>%
+   mutate(sa4_code = str_extract(labour_market_region_sa4_asgs_2011, "^\\d+"), year = as.integer(year)) %>%
+   filter(year <= 2020) %>%
+   select(sa4 = sa4_code, year, unemployment_rate)
> controls_all <- employment_panel %>%
+   full_join(income_clean, by = c("sa4", "year")) %>%
+   full_join(age_clean, by = c("sa4", "year")) %>%
+   full_join(population_clean %>% filter(year <= 2020), by = c("sa4", "year")) %>%
+   full_join(unemployment_clean, by = c("sa4", "year")) %>%
+   mutate(employment_rate = employed / population)
> controls_avg <- controls_all %>%
+   group_by(sa4) %>%
+   summarise(
+     employment_rate_avg   = mean(employment_rate, na.rm = TRUE),
+     log_med_income_avg    = mean(log_med_income, na.rm = TRUE),
+     age_25_64_avg         = mean(age_25_64, na.rm = TRUE),
+     unemployment_rate_avg = mean(unemployment_rate, na.rm = TRUE),
+     .groups = "drop"
+   )
>
> # PART 7b
> specialisation_2016 <- read_csv(f_specialisation_2016, show_col_types = FALSE) %>%
+   transmute(sa4 = as.character(sa4), specialisation_index = specialisation_index_2016)
>
> # PART 7c
> business_churn <- tibble(sa4 = character(0), business_churn = numeric(0))
>
> # PART 8 - NATIVE PAIRING: SE1 only in homeostasis_analysis, SE2 only in finance_analysis
> homeostasis_analysis <- homeostasis_cross %>%
+   left_join(se1_avg, by = "sa4") %>%
+   left_join(density_avg, by = "sa4") %>%
+   left_join(entropy_2016, by = "sa4") %>%
+   left_join(ccs_2016, by = "sa4") %>%
+   left_join(controls_avg, by = "sa4") %>%
+   left_join(network_size_migration, by = "sa4") %>%
+   left_join(business_churn, by = "sa4") %>%
+   left_join(specialisation_2016, by = "sa4")
> finance_analysis <- new_firms_percapita %>%
+   left_join(se2_avg, by = "sa4") %>%
+   mutate(se2_avg = replace_na(se2_avg, 0)) %>%
+   left_join(density_avg, by = "sa4") %>%
+   left_join(entropy_2016, by = "sa4") %>%
+   left_join(ccs_2016, by = "sa4") %>%
```

```
+   left_join(controls_avg, by = "sa4") %>%
+   left_join(burt_constraint, by = "sa4") %>%
+   mutate(burt_constraint = replace_na(burt_constraint, 0)) %>%
+   left_join(in_deals_pc_final, by = "sa4") %>%
+   mutate(in_deals_pc = replace_na(in_deals_pc, 0)) %>%
+   left_join(network_size_vc, by = "sa4") %>%
+   mutate(network_size_vc = replace_na(network_size_vc, 0)) %>%
+   left_join(business_churn, by = "sa4") %>%
+   left_join(specialisation_2016, by = "sa4")
> homeostasis_analysis <- homeostasis_analysis %>% mutate(density_100 = population_density_avg / 100)
> finance_analysis <- finance_analysis %>% mutate(density_100 = population_density_avg / 100)
> homeostasis_analysis <- homeostasis_analysis %>%
+   mutate(se1_c = se1_avg - mean(se1_avg, na.rm = TRUE),
+          ccs_c = creative_class_share - mean(creative_class_share, na.rm = TRUE),
+          entropy_c = entropy - mean(entropy, na.rm = TRUE))
> finance_analysis <- finance_analysis %>%
+   mutate(se2_c = se2_avg - mean(se2_avg, na.rm = TRUE),
+          ccs_c = creative_class_share - mean(creative_class_share, na.rm = TRUE),
+          entropy_c = entropy - mean(entropy, na.rm = TRUE))
> homeostasis_analysis <- homeostasis_analysis %>%
+   mutate(z_homeostasis = zscore(homeostasis), z_se1 = zscore(se1_avg),
+          z_ccs = zscore(creative_class_share), z_entropy = zscore(entropy),
+          z_density = zscore(population_density_avg),
+          z_income = zscore(log_med_income_avg), z_unemp = zscore(unemployment_rate_avg),
+          z_age = zscore(age_25_64_avg), z_emprate = zscore(employment_rate_avg),
+          z_specialisation = zscore(specialisation_index))
> finance_analysis <- finance_analysis %>%
+   mutate(z_new_firms = zscore(new_firms_percapita), z_se2 = zscore(se2_avg),
+          z_ccs = zscore(creative_class_share), z_entropy = zscore(entropy),
+          z_density = zscore(population_density_avg),
+          z_income = zscore(log_med_income_avg), z_unemp = zscore(unemployment_rate_avg),
+          z_age = zscore(age_25_64_avg), z_emprate = zscore(employment_rate_avg),
+          z_burtc = zscore(burt_constraint), z_indeals = zscore(in_deals_pc),
+          z_specialisation = zscore(specialisation_index))
> cat(sprintf("[PART 8] homeostasis_analysis: %d rows | finance_analysis: %d rows\n",
+             nrow(homeostasis_analysis), nrow(finance_analysis)))
[PART 8] homeostasis_analysis: 88 rows | finance_analysis: 89 rows
> vif_check_homeo <- lm(homeostasis ~ se1_c + ccs_c + entropy_c +
+                         log_med_income_avg + unemployment_rate_avg + age_25_64_avg +
+                         employment_rate_avg + density_100 + specialisation_index,
+                       data = homeostasis_analysis)
> homeo_vif <- vif(vif_check_homeo)
> cat(sprintf("[PART 8] MEAN VIF (Homeostasis): %.3f\n", mean(homeo_vif)))
[PART 8] MEAN VIF (Homeostasis): 4.215
> vif_check_finance <- lm(new_firms_percapita ~ se2_c + ccs_c + entropy_c +
+                           log_med_income_avg + unemployment_rate_avg + age_25_64_avg +
+                           employment_rate_avg + density_100 + burt_constraint + in_deals_pc + specialisation_index,
+                         data = finance_analysis)
> fin_vif <- vif(vif_check_finance)
> cat(sprintf("[PART 8] MEAN VIF (Finance): %.3f\n", mean(fin_vif)))
[PART 8] MEAN VIF (Finance): 4.172
>
> # PART 9 - NATIVE PAIRING Model 4 (SE1 only, no SE2)
> controls_formula_base   <- "density_100 + log_med_income_avg + unemployment_rate_avg + age_25_64_avg + employment_rate_avg + specialisation_index"
> controls_formula_base_z <- "z_density + z_income + z_unemp + z_age + z_emprate + z_specialisation"
> f_h1 <- as.formula(paste("homeostasis ~ se1_c +", controls_formula_base))
> f_h2 <- as.formula(paste("homeostasis ~ ccs_c +", controls_formula_base))
> f_h3 <- as.formula(paste("homeostasis ~ entropy_c +", controls_formula_base))
> f_h4 <- as.formula(paste("homeostasis ~ se1_c + ccs_c + entropy_c +", controls_formula_base))
> f_h5 <- as.formula(paste("homeostasis ~ se1_c * entropy_c + ccs_c +", controls_formula_base))
> f_h6 <- as.formula(paste("homeostasis ~ entropy_c * ccs_c + se1_c +", controls_formula_base))
> f_h7 <- as.formula(paste("homeostasis ~ se1_c * ccs_c + entropy_c +", controls_formula_base))
> f_h8 <- as.formula(paste("homeostasis ~ se1_c * entropy_c * ccs_c +", controls_formula_base))
> f_h1z <- as.formula(paste("z_homeostasis ~ z_se1 +", controls_formula_base_z))
> f_h2z <- as.formula(paste("z_homeostasis ~ z_ccs +", controls_formula_base_z))
> f_h3z <- as.formula(paste("z_homeostasis ~ z_entropy +", controls_formula_base_z))
> f_h4z <- as.formula(paste("z_homeostasis ~ z_se1 + z_ccs + z_entropy +", controls_formula_base_z))
> f_h5z <- as.formula(paste("z_homeostasis ~ z_se1 * z_entropy + z_ccs +", controls_formula_base_z))
> f_h6z <- as.formula(paste("z_homeostasis ~ z_entropy * z_ccs + z_se1 +", controls_formula_base_z))
> f_h7z <- as.formula(paste("z_homeostasis ~ z_se1 * z_ccs + z_entropy +", controls_formula_base_z))
> f_h8z <- as.formula(paste("z_homeostasis ~ z_se1 * z_entropy * z_ccs +", controls_formula_base_z))
> homeo_raw <- lapply(list(f_h1, f_h2, f_h3, f_h4, f_h5, f_h6, f_h7, f_h8), lm, data = homeostasis_analysis)
> homeo_std <- lapply(list(f_h1z, f_h2z, f_h3z, f_h4z, f_h5z, f_h6z, f_h7z, f_h8z), lm, data = homeostasis_analysis)
> homeo_var_specs <- list(
+   "SE1 (Migration Emb., <=2020)"   = list(raw = "se1_c",                   z = "z_se1"),
```

```
+   "Creative Class Share (2016)"    = list(raw = "ccs_c",                    z = "z_ccs"),
+   "Diversity (2016)"                = list(raw = "entropy_c",                z = "z_entropy"),
+   "SE1 x Diversity"                 = list(raw = "se1_c:entropy_c",          z = "z_se1:z_entropy"),
+   "Diversity x Creative Class"      = list(raw = "entropy_c:ccs_c",          z = "z_entropy:z_ccs"),
+   "SE1 x Creative Class"            = list(raw = "se1_c:ccs_c",              z = "z_se1:z_ccs"),
+   "SE1 x Diversity x Creative Class" = list(raw = "se1_c:entropy_c:ccs_c",   z = "z_se1:z_entropy:z_ccs"),
+   "Density (<=2020, control)"       = list(raw = "density_100",              z = "z_density"),
+   "Median Income (log, <=2020)"     = list(raw = "log_med_income_avg",       z = "z_income"),
+   "Unemployment (<=2020)"           = list(raw = "unemployment_rate_avg",    z = "z_unemp"),
+   "Demographics (<=2020)"           = list(raw = "age_25_64_avg",            z = "z_age"),
+   "Employment Rate (<=2020)"        = list(raw = "employment_rate_avg",      z = "z_emprate"),
+   "Specialisation Index (2016)"     = list(raw = "specialisation_index",     z = "z_specialisation")
+ )
> cat("\n\nTABLE 3A-LAGGED (NATIVE PAIRING).\n")


TABLE 3A-LAGGED (NATIVE PAIRING).
> print_results_table(homeo_raw, homeo_std, c("(1)","(2)","(3)","(4)","(5)","(6)","(7)","(8)"), homeo_var_specs)
SE1 (Migration Emb., <=2020)    -6.127 |  |  | -14.505** | -14.533** | -14.869** | -14.423** | -20.375**
                                (6.174) |  |  | (4.994) | (5.061) | (4.953) | (5.105) | (6.930)
                                [-0.106] |  |  | [-0.251] | [-0.251] | [-0.257] | [-0.249] | [-0.352]
Creative Class Share (2016)      | 35.465+ |  | 28.471* | 28.520* | 28.707* | 28.159* | 31.574**
                                 | (18.467) |  | (12.505) | (12.350) | (13.677) | (12.287) | (9.545)
                                 | [0.580] |  | [0.465] | [0.466] | [0.469] | [0.460] | [0.516]
Diversity (2016)                 |  | 0.083*** | 0.086*** | 0.086*** | 0.087*** | 0.086*** | 0.081***
                                 |  | (0.013) | (0.014) | (0.013) | (0.014) | (0.013) | (0.012)
                                 |  | [0.585] | [0.609] | [0.607] | [0.614] | [0.609] | [0.571]
SE1 x Diversity                  |  |  |  | 0.049 |  |  | 0.718
                                 |  |  |  | (0.459) |  |  | (0.620)
                                 |  |  |  | [0.012] |  |  | [0.183]
Diversity x Creative Class       |  |  |  |  | -0.413 |  | -0.566
                                 |  |  |  |  | (0.622) |  | (0.444)
                                 |  |  |  |  | [-0.099] |  | [-0.136]
SE1 x Creative Class             |  |  |  |  |  | -192.485 | -442.935+
                                 |  |  |  |  |  | (143.254) | (223.323)
                                 |  |  |  |  |  | [-0.114] | [-0.262]
SE1 x Diversity x Creative Class |  |  |  |  |  |  | 25.639+
                                 |  |  |  |  |  |  | (14.558)
                                 |  |  |  |  |  |  | [0.223]
Density (<=2020, control)       0.102*** | 0.051 | 0.064*** | 0.026 | 0.026 | 0.034+ | 0.027 | 0.033+
                                (0.017) | (0.031) | (0.018) | (0.022) | (0.022) | (0.019) | (0.023) | (0.019)
                                [0.618] | [0.306] | [0.387] | [0.154] | [0.154] | [0.207] | [0.164] | [0.202]
Median Income (log, <=2020)     1.167 | -4.056 | -0.828 | -3.960+ | -3.955+ | -3.923+ | -3.769+ | -4.310*
                                (1.339) | (2.889) | (1.107) | (2.026) | (2.041) | (2.114) | (2.007) | (1.758)
                                [0.121] | [-0.422] | [-0.086] | [-0.412] | [-0.412] | [-0.408] | [-0.392] | [-0.448]
Unemployment (<=2020)           -0.185 | -0.300 | -0.409* | -0.448** | -0.446** | -0.443** | -0.468** | -0.482**
                                (0.202) | (0.185) | (0.155) | (0.152) | (0.155) | (0.150) | (0.149) | (0.148)
                                [-0.110] | [-0.178] | [-0.243] | [-0.266] | [-0.265] | [-0.263] | [-0.278] | [-0.286]
Demographics (<=2020)           -0.089 | 0.089 | -0.170+ | -0.043 | -0.042 | -0.063 | -0.061 | -0.040
                                (0.094) | (0.127) | (0.088) | (0.103) | (0.104) | (0.103) | (0.107) | (0.105)
                                [-0.140] | [0.141] | [-0.269] | [-0.068] | [-0.066] | [-0.100] | [-0.096] | [-0.063]
Employment Rate (<=2020)        4.759 | 2.880 | 4.305 | 2.380 | 2.400 | 2.736 | 2.205 | 2.537
                                (3.700) | (3.403) | (3.279) | (2.916) | (2.953) | (2.940) | (2.865) | (2.830)
                                [0.155] | [0.094] | [0.140] | [0.077] | [0.078] | [0.089] | [0.072] | [0.083]
Specialisation Index (2016)     0.031 | 0.070+ | 0.113** | 0.116*** | 0.116*** | 0.134** | 0.116*** | 0.145***
                                (0.043) | (0.039) | (0.038) | (0.032) | (0.032) | (0.048) | (0.032) | (0.042)
                                [0.076] | [0.172] | [0.279] | [0.285] | [0.286] | [0.331] | [0.285] | [0.359]
-------------------------------------------------------------------------------------------------------
Observations                    80 | 80 | 80 | 80 | 80 | 80 | 80 | 80
Adjusted R-squared              0.524 | 0.569 | 0.672 | 0.729 | 0.725 | 0.728 | 0.731 | 0.736
AIC                             297.2 | 289.3 | 267.4 | 253.9 | 255.9 | 255.2 | 254.2 | 255.0

Model columns: (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8)
Robust (HC1) SE in parentheses (); standardized coefficient in brackets [].
*** p<0.001  ** p<0.01  * p<0.05  + p<0.1

>
> # PART 10 - NATIVE PAIRING Model 4 (SE2 only, no SE1)
> controls_formula_fin   <- paste(controls_formula_base, "+ burt_constraint + in_deals_pc")
> controls_formula_fin_z <- paste(controls_formula_base_z, "+ z_burtc + z_indeals")
> f_f1 <- as.formula(paste("new_firms_percapita ~ se2_c +", controls_formula_fin))
> f_f2 <- as.formula(paste("new_firms_percapita ~ ccs_c +", controls_formula_fin))
> f_f3 <- as.formula(paste("new_firms_percapita ~ entropy_c +", controls_formula_fin))
> f_f4 <- as.formula(paste("new_firms_percapita ~ se2_c + ccs_c + entropy_c +", controls_formula_fin))
> f_f5 <- as.formula(paste("new_firms_percapita ~ se2_c * entropy_c + ccs_c +", controls_formula_fin))
```

```
> f_f6 <- as.formula(paste("new_firms_percapita ~ entropy_c * ccs_c + se2_c +", controls_formula_fin))
> f_f7 <- as.formula(paste("new_firms_percapita ~ se2_c * ccs_c + entropy_c +", controls_formula_fin))
> f_f8 <- as.formula(paste("new_firms_percapita ~ se2_c * entropy_c * ccs_c +", controls_formula_fin))
> f_f1z <- as.formula(paste("z_new_firms ~ z_se2 +", controls_formula_fin_z))
> f_f2z <- as.formula(paste("z_new_firms ~ z_ccs +", controls_formula_fin_z))
> f_f3z <- as.formula(paste("z_new_firms ~ z_entropy +", controls_formula_fin_z))
> f_f4z <- as.formula(paste("z_new_firms ~ z_se2 + z_ccs + z_entropy +", controls_formula_fin_z))
> f_f5z <- as.formula(paste("z_new_firms ~ z_se2 * z_entropy + z_ccs +", controls_formula_fin_z))
> f_f6z <- as.formula(paste("z_new_firms ~ z_entropy * z_ccs + z_se2 +", controls_formula_fin_z))
> f_f7z <- as.formula(paste("z_new_firms ~ z_se2 * z_ccs + z_entropy +", controls_formula_fin_z))
> f_f8z <- as.formula(paste("z_new_firms ~ z_se2 * z_entropy * z_ccs +", controls_formula_fin_z))
> fin_raw <- lapply(list(f_f1, f_f2, f_f3, f_f4, f_f5, f_f6, f_f7, f_f8), lm, data = finance_analysis)
> fin_std <- lapply(list(f_f1z, f_f2z, f_f3z, f_f4z, f_f5z, f_f6z, f_f7z, f_f8z), lm, data = finance_analysis)
> fin_var_specs <- list(
+   "SE2 (Finance Emb., <=2020)"       = list(raw = "se2_c",                   z = "z_se2"),
+   "Creative Class Share (2016)"      = list(raw = "ccs_c",                   z = "z_ccs"),
+   "Diversity (2016)"                  = list(raw = "entropy_c",               z = "z_entropy"),
+   "SE2 x Diversity"                   = list(raw = "se2_c:entropy_c",         z = "z_se2:z_entropy"),
+   "Diversity x Creative Class"        = list(raw = "entropy_c:ccs_c",         z = "z_entropy:z_ccs"),
+   "SE2 x Creative Class"              = list(raw = "se2_c:ccs_c",             z = "z_se2:z_ccs"),
+   "SE2 x Diversity x Creative Class"  = list(raw = "se2_c:entropy_c:ccs_c",    z = "z_se2:z_entropy:z_ccs"),
+   "Density (<=2020, control)"         = list(raw = "density_100",             z = "z_density"),
+   "Median Income (log, <=2020)"       = list(raw = "log_med_income_avg",      z = "z_income"),
+   "Unemployment (<=2020)"             = list(raw = "unemployment_rate_avg",   z = "z_unemp"),
+   "Demographics (<=2020)"             = list(raw = "age_25_64_avg",           z = "z_age"),
+   "Employment Rate (<=2020)"          = list(raw = "employment_rate_avg",     z = "z_emprate"),
+   "Burt's Constraint (VC, <=2020)"    = list(raw = "burt_constraint",         z = "z_burtc"),
+   "VC Deal Volume, per capita (<=2020)" = list(raw = "in_deals_pc",           z = "z_indeals"),
+   "Specialisation Index (2016)"       = list(raw = "specialisation_index",    z = "z_specialisation")
+ )
> cat("\n\nTABLE 3B-LAGGED (NATIVE PAIRING).\n")


TABLE 3B-LAGGED (NATIVE PAIRING).
> print_results_table(fin_raw, fin_std, c("(1)","(2)","(3)","(4)","(5)","(6)","(7)","(8)"), fin_var_specs)
SE2 (Finance Emb., <=2020)      -0.033** |  |  | -0.031* | -0.030* | -0.031* | -0.030* | -0.008
                                (0.011) |  |  | (0.012) | (0.012) | (0.012) | (0.012) | (0.010)
                                [-0.163] |  |  | [-0.153] | [-0.146] | [-0.152] | [-0.147] | [-0.039]
Creative Class Share (2016)      | 0.108 |  | 0.260 | 0.284 | 0.259 | 0.264 | 0.457*
                                 | (0.197) |  | (0.194) | (0.198) | (0.194) | (0.208) | (0.181)
                                 | [0.047] |  | [0.113] | [0.124] | [0.113] | [0.115] | [0.199]
Diversity (2016)                 |  | -0.001+ | -0.000 | -0.000 | -0.000 | -0.000 | -0.001
                                 |  | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000)
                                 |  | [-0.121] | [-0.086] | [-0.068] | [-0.086] | [-0.087] | [-0.110]
SE2 x Diversity                  |  |  |  | -0.000 |  |  | -0.000
                                 |  |  |  | (0.001) |  |  | (0.001)
                                 |  |  |  | [-0.031] |  |  | [-0.021]
Diversity x Creative Class       |  |  |  |  | 0.001 |  | 0.007
                                 |  |  |  |  | (0.017) |  | (0.014)
                                 |  |  |  |  | [0.007] |  | [0.042]
SE2 x Creative Class             |  |  |  |  |  | -0.427 | 0.234
                                 |  |  |  |  |  | (0.536) | (0.578)
                                 |  |  |  |  |  | [-0.072] | [0.040]
SE2 x Diversity x Creative Class |  |  |  |  |  |  | -0.121***
                                 |  |  |  |  |  |  | (0.033)
                                 |  |  |  |  |  |  | [-0.301]
Density (<=2020, control)       0.001 | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 | 0.001 | 0.001
                                (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001)
                                [0.098] | [0.063] | [0.137] | [0.078] | [0.069] | [0.074] | [0.093] | [0.102]
Median Income (log, <=2020)     -0.097* | -0.108* | -0.083+ | -0.124* | -0.128* | -0.124* | -0.115* | -0.124*
                                (0.041) | (0.050) | (0.043) | (0.049) | (0.051) | (0.050) | (0.051) | (0.051)
                                [-0.270] | [-0.300] | [-0.230] | [-0.343] | [-0.356] | [-0.343] | [-0.318] | [-0.345]
Unemployment (<=2020)           -0.000 | -0.001 | 0.001 | 0.000 | 0.001 | 0.000 | 0.000 | -0.002
                                (0.003) | (0.004) | (0.004) | (0.003) | (0.003) | (0.003) | (0.003) | (0.003)
                                [-0.003] | [-0.016] | [0.012] | [0.005] | [0.014] | [0.005] | [0.002] | [-0.026]
Demographics (<=2020)           0.011*** | 0.012** | 0.012** | 0.013*** | 0.013*** | 0.013*** | 0.012** | 0.012***
                                (0.003) | (0.004) | (0.004) | (0.003) | (0.003) | (0.004) | (0.004) | (0.003)
                                [0.465] | [0.532] | [0.533] | [0.545] | [0.546] | [0.548] | [0.519] | [0.524]
Employment Rate (<=2020)        0.013 | 0.023 | 0.037 | 0.014 | 0.017 | 0.013 | 0.009 | -0.006
                                (0.061) | (0.062) | (0.060) | (0.059) | (0.060) | (0.063) | (0.058) | (0.059)
                                [0.011] | [0.020] | [0.032] | [0.012] | [0.015] | [0.011] | [0.008] | [-0.005]
Burt's Constraint (VC, <=2020)  -0.045*** | -0.060*** | -0.055*** | -0.043*** | -0.045*** | -0.044*** | -0.045*** | -0.043***
                                (0.011) | (0.014) | (0.013) | (0.011) | (0.012) | (0.011) | (0.011) | (0.010)
                                [-0.230] | [-0.309] | [-0.282] | [-0.224] | [-0.231] | [-0.224] | [-0.229] | [-0.221]
VC Deal Volume, per capita (<=2020)3.707*** | 3.391*** | 3.348*** | 3.532*** | 3.496*** | 3.531*** | 3.418*** | 3.263***
```

```
                              (0.765) | (0.755) | (0.726) | (0.779) | (0.764) | (0.783) | (0.793) | (0.688)
                              [0.727] | [0.666] | [0.657] | [0.693] | [0.686] | [0.693] | [0.671] | [0.640]
Specialisation Index (2016)   -0.001 | -0.001 | -0.001 | -0.001 | -0.001 | -0.001 | -0.001 | -0.001
                              (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001)
                              [-0.056] | [-0.042] | [-0.071] | [-0.054] | [-0.042] | [-0.058] | [-0.041] | [-0.089]
-------------------------------------------------------------------------------------------------
Observations                  80 | 80 | 80 | 80 | 80 | 80 | 80 | 80
Adjusted R-squared            0.850 | 0.829 | 0.835 | 0.850 | 0.848 | 0.847 | 0.850 | 0.873
AIC                           -317.6 | -307.1 | -309.9 | -315.6 | -314.0 | -313.6 | -314.7 | -325.9

Model columns: (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8)
Robust (HC1) SE in parentheses (); standardized coefficient in brackets [].
*** p<0.001  ** p<0.01  * p<0.05  + p<0.1

> cat("\n=== DONE ===\n")

=== DONE ===
> desc_vars_homeo <- homeostasis_analysis %>%
+   filter(complete.cases(select(., homeostasis, se1_c, ccs_c, entropy_c, density_100,
+                                log_med_income_avg, unemployment_rate_avg, age_25_64_avg,
+                                employment_rate_avg, specialisation_index)))
> desc_vars_fin <- finance_analysis %>%
+   filter(complete.cases(select(., new_firms_percapita, se2_c, ccs_c, entropy_c, density_100,
+                                log_med_income_avg, unemployment_rate_avg, age_25_64_avg,
+                                employment_rate_avg, burt_constraint, in_deals_pc, specialisation_index)))
>
> cat("N homeo:", nrow(desc_vars_homeo), "| N finance:", nrow(desc_vars_fin), "\n\n")
N homeo: 80 | N finance: 80

>
> summary_table <- function(df, vars) {
+   t(sapply(df[vars], function(x) c(N=sum(!is.na(x)), Mean=mean(x,na.rm=T), SD=sd(x,na.rm=T), Min=min(x,na.rm=T), Max=max(x,na.rm=T))))
+ }
>
> print(round(summary_table(desc_vars_homeo, c("homeostasis","se1_c","ccs_c","entropy_c","density_100",
+                                              "log_med_income_avg","unemployment_rate_avg","age_25_64_avg",
+                                              "employment_rate_avg","specialisation_index")), 3))
                       N   Mean     SD     Min    Max
homeostasis           80  0.134  2.117  -3.477  6.442
se1_c                 80  0.000  0.030  -0.046  0.078
ccs_c                 80  0.002  0.034  -0.048  0.102
entropy_c             80  0.882 14.908 -18.122 33.882
density_100           80  7.475 12.997   0.001 52.725
log_med_income_avg    80  6.763  0.209   6.397  7.296
unemployment_rate_avg 80  5.702  1.244   3.173  9.188
age_25_64_avg         80 52.234  3.221  46.350 65.213
employment_rate_avg   80  0.475  0.068   0.142  0.600
specialisation_index  80 12.591  4.744   5.638 29.162
>
> print(round(summary_table(desc_vars_fin, c("new_firms_percapita","se2_c","burt_constraint","in_deals_pc")), 3))
                     N  Mean    SD    Min   Max
new_firms_percapita 80 0.028 0.080  0.000 0.622
se2_c               80 0.017 0.392 -0.262 0.738
burt_constraint     80 0.430 0.401  0.000 1.000
in_deals_pc         80 0.010 0.016  0.000 0.093
> # Homeostasis-side correlations (se1_c is native here; se2_c doesn't exist in this dataset)
> cor_vars_homeo <- desc_vars_homeo %>%
+   select(homeostasis, se1_c, ccs_c, entropy_c, density_100, log_med_income_avg, age_25_64_avg)
> print(round(cor(cor_vars_homeo, use = "pairwise.complete.obs"), 3))
                  homeostasis se1_c ccs_c entropy_c density_100 log_med_income_avg age_25_64_avg
homeostasis             1.000 0.050 0.676     0.654       0.705             0.539         0.509
se1_c                   0.050 1.000 0.321     0.391       0.185             0.274         0.204
ccs_c                   0.676 0.321 1.000     0.617       0.729             0.830         0.582
entropy_c               0.654 0.391 0.617     1.000       0.626             0.550         0.634
density_100             0.705 0.185 0.729     0.626       1.000             0.616         0.732
log_med_income_avg      0.539 0.274 0.830     0.550       0.616             1.000         0.748
age_25_64_avg           0.509 0.204 0.582     0.634       0.732             0.748         1.000
>
> # Finance-side correlations (se2_c is native here; se1_c doesn't exist in this dataset)
> cor_vars_fin <- desc_vars_fin %>%
+   select(new_firms_percapita, se2_c, ccs_c, entropy_c)
> print(round(cor(cor_vars_fin, use = "pairwise.complete.obs"), 3))
                    new_firms_percapita  se2_c ccs_c entropy_c
new_firms_percapita               1.000 -0.051 0.457     0.278
se2_c                            -0.051  1.000 0.197     0.290
```

```
ccs_c                           0.457  0.197 1.000     0.617
entropy_c                       0.278  0.290 0.617     1.000
> combined_check <- homeostasis_analysis %>%
+   select(sa4, se1_c) %>%
+   inner_join(finance_analysis %>% select(sa4, se2_c), by = "sa4")
> cat("N for SC-SE correlation:", nrow(combined_check), "\n")
N for SC-SE correlation: 88
> print(round(cor(combined_check$se1_c, combined_check$se2_c, use = "pairwise.complete.obs"), 3))
[1] 0.125
> # Table 3C: SE alone on Homeostasis (requires se2_avg joined into homeostasis_analysis)
> homeostasis_analysis_c <- homeostasis_analysis %>% left_join(se2_avg, by="sa4") %>%
+   mutate(se2_avg = replace_na(se2_avg, 0), se2_c = se2_avg - mean(se2_avg, na.rm=TRUE))
> # then run f_h1...f_h8 equivalents with se2_c in place of se1_c
>
> # Table 3D: SC alone on New Firm Formation (requires se1_avg joined into finance_analysis)
> finance_analysis_d <- finance_analysis %>% left_join(se1_avg, by="sa4") %>%
+   mutate(se1_c = se1_avg - mean(se1_avg, na.rm=TRUE))
> # then run f_f1...f_f8 equivalents with se1_c in place of se2_c
> desc_vars_homeo <- homeostasis_analysis %>%
+   filter(complete.cases(select(., homeostasis, se1_c, ccs_c, entropy_c, density_100,
+                                log_med_income_avg, unemployment_rate_avg, age_25_64_avg, employment_rate_avg, specialisation_index)))
> desc_vars_fin <- finance_analysis %>%
+   filter(complete.cases(select(., new_firms_percapita, se2_c, ccs_c, entropy_c, density_100,
+                                log_med_income_avg, unemployment_rate_avg, age_25_64_avg, employment_rate_avg,
+                                burt_constraint, in_deals_pc, specialisation_index)))
> cat("N homeo:", nrow(desc_vars_homeo), "| N finance:", nrow(desc_vars_fin), "\n")
N homeo: 80 | N finance: 80
> library(car)
> h <- desc_vars_homeo; f <- desc_vars_fin
> vh <- vif(lm(homeostasis ~ se1_c + ccs_c + entropy_c + log_med_income_avg + unemployment_rate_avg + age_25_64_avg + employment_rate_avg + density_100 + specialisation_index, data = h))
> vf <- vif(lm(new_firms_percapita ~ se2_c + ccs_c + entropy_c + log_med_income_avg + unemployment_rate_avg + age_25_64_avg + employment_rate_avg + density_100 + burt_constraint + in_deals_pc +
    specialisation_index, data = f))
> print(round(vh,2)); cat("Mean VIF homeo:", round(mean(vh),3), "\n")
               se1_c                 ccs_c             entropy_c    log_med_income_avg unemployment_rate_avg         age_25_64_avg   employment_rate_avg           density_100  specialisation_index
                1.26                  7.13                  2.62                  8.32                  2.65                  5.92                  3.44                  4.51                  2.09
Mean VIF homeo: 4.215
> print(round(vf,2)); cat("Mean VIF finance:", round(mean(vf),3), "\n")
               se2_c                 ccs_c             entropy_c    log_med_income_avg unemployment_rate_avg         age_25_64_avg   employment_rate_avg           density_100       burt_constraint
                1.46                  8.38                  2.91                  8.83                  2.66                  6.81                  3.57                  4.65                  1.82
         in_deals_pc  specialisation_index
                2.41                  2.39
Mean VIF finance: 4.172
> print(round(cor(h[,c("homeostasis","se1_c","ccs_c","entropy_c","density_100","log_med_income_avg","age_25_64_avg","employment_rate_avg","unemployment_rate_avg","specialisation_index")]),3))
                    homeostasis  se1_c  ccs_c entropy_c density_100 log_med_income_avg age_25_64_avg employment_rate_avg unemployment_rate_avg specialisation_index
homeostasis               1.000  0.050  0.676     0.654       0.705              0.539         0.509              0.498               -0.407                0.193
se1_c                     0.050  1.000  0.321     0.391       0.185              0.274         0.204              0.202                0.000               -0.203
ccs_c                     0.676  0.321  1.000     0.617       0.729              0.830         0.582              0.635               -0.420               -0.009
entropy_c                 0.654  0.391  0.617     1.000       0.626              0.550         0.634              0.466               -0.029               -0.161
density_100               0.705  0.185  0.729     0.626       1.000              0.616         0.732              0.506               -0.263                0.200
log_med_income_avg        0.539  0.274  0.830     0.550       0.616              1.000         0.748              0.618               -0.502                0.167
age_25_64_avg             0.509  0.204  0.582     0.634       0.732              0.748         1.000              0.528               -0.178                0.208
employment_rate_avg       0.498  0.202  0.635     0.466       0.506              0.618         0.528              1.000               -0.548               -0.228
unemployment_rate_avg    -0.407  0.000 -0.420    -0.029      -0.263             -0.502        -0.178             -0.548                1.000               -0.226
specialisation_index      0.193 -0.203 -0.009    -0.161       0.200              0.167         0.208             -0.228               -0.226                1.000
> print(round(cor(f[,c("new_firms_percapita","se2_c","ccs_c","entropy_c","burt_constraint","in_deals_pc")]),3))
                  new_firms_percapita  se2_c ccs_c entropy_c burt_constraint in_deals_pc
new_firms_percapita              1.000 -0.051 0.457     0.278          -0.049       0.832
se2_c                           -0.051  1.000 0.197     0.290           0.437       0.240
ccs_c                            0.457  0.197 1.000     0.617           0.370       0.591
entropy_c                        0.278  0.290 0.617     1.000           0.483       0.333
burt_constraint                 -0.049  0.437 0.370     0.483           1.000       0.198
in_deals_pc                      0.832  0.240 0.591     0.333           0.198       1.000
> vars <- c("homeostasis","new_firms_percapita","se1_c","se2_c","ccs_c","entropy_c","density_100","log_med_income_avg","unemployment_rate_avg","age_25_64_avg","employment_rate_avg","specialisation_index","
    burt_constraint","in_deals_pc")
> common <- homeostasis_analysis %>%
+   select(sa4, homeostasis, se1_c, ccs_c, entropy_c, density_100, log_med_income_avg, unemployment_rate_avg, age_25_64_avg, employment_rate_avg, specialisation_index) %>%
+   inner_join(finance_analysis %>% select(sa4, new_firms_percapita, se2_c, burt_constraint, in_deals_pc), by = "sa4") %>%
+   filter(complete.cases(across(all_of(vars))))
> cat("Common N:", nrow(common), "\n")
Common N: 80
> cat("Same 80 regions in both models? ", setequal(desc_vars_homeo$sa4, desc_vars_fin$sa4), "\n")
Same 80 regions in both models?  TRUE
> cat("Homeo-only regions:", length(setdiff(desc_vars_homeo$sa4, desc_vars_fin$sa4)), "| Fin-only regions:", length(setdiff(desc_vars_fin$sa4, desc_vars_homeo$sa4)), "\n")
Homeo-only regions: 0 | Fin-only regions: 0
> print(round(t(sapply(common[vars], function(x) c(N=length(x), Mean=mean(x), SD=sd(x), Min=min(x), Max=max(x)))), 3))
```

```
                       N   Mean     SD     Min    Max
homeostasis           80  0.134  2.117  -3.477  6.442
new_firms_percapita   80  0.028  0.080   0.000  0.622
se1_c                 80  0.000  0.030  -0.046  0.078
se2_c                 80  0.017  0.392  -0.262  0.738
ccs_c                 80  0.002  0.034  -0.048  0.102
entropy_c             80  0.882 14.908 -18.122 33.882
density_100           80  7.475 12.997   0.001 52.725
log_med_income_avg    80  6.763  0.209   6.397  7.296
unemployment_rate_avg 80  5.702  1.244   3.173  9.188
age_25_64_avg         80 52.234  3.221  46.350 65.213
employment_rate_avg   80  0.475  0.068   0.142  0.600
specialisation_index  80 12.591  4.744   5.638 29.162
burt_constraint       80  0.430  0.401   0.000  1.000
in_deals_pc           80  0.010  0.016   0.000  0.093
> cm <- cor(common[vars]); print(round(cm, 3))
                      homeostasis new_firms_percapita  se1_c  se2_c  ccs_c entropy_c density_100 log_med_income_avg unemployment_rate_avg age_25_64_avg employment_rate_avg specialisation_index
homeostasis                 1.000               0.519  0.050  0.277  0.676     0.654       0.705              0.539                -0.407         0.509              0.498                0.193
new_firms_percapita         0.519               1.000  0.069 -0.051  0.457     0.278       0.624              0.451                -0.180         0.676              0.397                0.302
se1_c                       0.050               0.069  1.000  0.203  0.321     0.391       0.185              0.274                 0.000         0.204              0.202               -0.203
se2_c                       0.277              -0.051  0.203  1.000  0.197     0.290       0.095              0.068                 0.066         0.065              0.098               -0.163
ccs_c                       0.676               0.457  0.321  0.197  1.000     0.617       0.729              0.830                -0.420         0.582              0.635               -0.009
entropy_c                   0.654               0.278  0.391  0.290  0.617     1.000       0.626              0.550                -0.029         0.634              0.466               -0.161
density_100                 0.705               0.624  0.185  0.095  0.729     0.626       1.000              0.616                -0.263         0.732              0.506                0.200
log_med_income_avg          0.539               0.451  0.274  0.068  0.830     0.550       0.616              1.000                -0.502         0.748              0.618                0.167
unemployment_rate_avg      -0.407              -0.180  0.000  0.066 -0.420    -0.029      -0.263             -0.502                 1.000        -0.178             -0.548               -0.226
age_25_64_avg               0.509               0.676  0.204  0.065  0.582     0.634       0.732              0.748                -0.178         1.000              0.528                0.208
employment_rate_avg         0.498               0.397  0.202  0.098  0.635     0.466       0.506              0.618                -0.548         0.528              1.000               -0.228
specialisation_index        0.193               0.302 -0.203 -0.163 -0.009    -0.161       0.200              0.167                -0.226         0.208             -0.228                1.000
burt_constraint             0.129              -0.049  0.337  0.437  0.370     0.483       0.209              0.336                -0.004         0.323              0.346               -0.357
in_deals_pc                 0.523               0.832  0.235  0.240  0.591     0.333       0.577              0.545                -0.264         0.601              0.463                0.243
                      burt_constraint in_deals_pc
homeostasis                     0.129       0.523
new_firms_percapita            -0.049       0.832
se1_c                           0.337       0.235
se2_c                           0.437       0.240
ccs_c                           0.370       0.591
entropy_c                       0.483       0.333
density_100                     0.209       0.577
log_med_income_avg              0.336       0.545
unemployment_rate_avg          -0.004      -0.264
age_25_64_avg                   0.323       0.601
employment_rate_avg             0.346       0.463
specialisation_index           -0.357       0.243
burt_constraint                 1.000       0.198
in_deals_pc                     0.198       1.000
> pm <- outer(vars, vars, Vectorize(function(a, b) if (a == b) 0 else cor.test(common[[a]], common[[b]])$p.value))
> dimnames(pm) <- dimnames(cm); print(signif(pm, 3))
                      homeostasis new_firms_percapita    se1_c    se2_c    ccs_c entropy_c density_100 log_med_income_avg unemployment_rate_avg age_25_64_avg employment_rate_avg specialisation_index
homeostasis             0.00e+00            8.18e-07 0.660000 1.28e-02 5.91e-12  4.80e-11    2.91e-13           2.56e-07             1.76e-04     1.43e-06           2.65e-06              0.08570
new_firms_percapita     8.18e-07            0.00e+00 0.542000 6.52e-01 1.99e-05  1.27e-02    6.46e-10           2.70e-05             1.10e-01     6.03e-12           2.62e-04              0.00648
se1_c                   6.60e-01            5.42e-01 0.000000 7.04e-02 3.65e-03  3.33e-04    1.00e-01           1.41e-02             9.97e-01     6.99e-02           7.19e-02              0.07030
se2_c                   1.28e-02            6.52e-01 0.070400 0.00e+00 8.00e-02  8.95e-03    4.01e-01           5.49e-01             5.61e-01     5.65e-01           3.89e-01              0.14700
ccs_c                   5.91e-12            1.99e-05 0.003650 8.00e-02 0.00e+00  1.13e-09    1.75e-14           1.94e-21             1.06e-04     1.54e-08           2.53e-10              0.93400
entropy_c               4.80e-11            1.27e-02 0.000333 8.95e-03 1.13e-09  0.00e+00    5.46e-10           1.23e-07             7.98e-01     2.78e-10           1.35e-05              0.15400
density_100             2.91e-13            6.46e-10 0.100000 4.01e-01 1.75e-14  5.46e-10    0.00e+00           1.21e-09             1.85e-02     1.15e-14           1.72e-06              0.07480
log_med_income_avg      2.56e-07            2.70e-05 0.014100 5.49e-01 1.94e-21  1.23e-07    1.21e-09           0.00e+00             2.14e-06     1.47e-15           9.80e-10              0.13900
unemployment_rate_avg   1.76e-04            1.10e-01 0.997000 5.61e-01 1.06e-04  7.98e-01    1.85e-02           2.14e-06             0.00e+00     1.14e-01           1.43e-07              0.04430
age_25_64_avg           1.43e-06            6.03e-12 0.069900 5.65e-01 1.54e-08  2.78e-10    1.15e-14           1.47e-15             1.14e-01     0.00e+00           4.90e-07              0.06360
employment_rate_avg     2.65e-06            2.62e-04 0.071900 3.89e-01 2.53e-10  1.35e-05    1.72e-06           9.80e-10             1.43e-07     4.90e-07           0.00e+00              0.04160
specialisation_index    8.57e-02            6.48e-03 0.070300 1.47e-01 9.34e-01  1.54e-01    7.48e-02           1.39e-01             4.43e-02     6.36e-02           4.16e-02              0.00000
burt_constraint         2.54e-01            6.64e-01 0.002210 5.16e-05 7.33e-04  5.69e-06    6.31e-02           2.29e-03             9.69e-01     3.49e-03           1.65e-03              0.00115
in_deals_pc             6.29e-07            1.15e-21 0.036200 3.20e-02 7.97e-09  2.52e-03    2.11e-08           1.70e-07             1.78e-02     3.89e-09           1.51e-05              0.02990
                      burt_constraint in_deals_pc
homeostasis                  2.54e-01    6.29e-07
new_firms_percapita          6.64e-01    1.15e-21
se1_c                        2.21e-03    3.62e-02
se2_c                        5.16e-05    3.20e-02
ccs_c                        7.33e-04    7.97e-09
entropy_c                    5.69e-06    2.52e-03
density_100                  6.31e-02    2.11e-08
log_med_income_avg           2.29e-03    1.70e-07
unemployment_rate_avg        9.69e-01    1.78e-02
age_25_64_avg                3.49e-03    3.89e-09
employment_rate_avg          1.65e-03    1.51e-05
```

```
specialisation_index         1.15e-03    2.99e-02
burt_constraint              0.00e+00    7.78e-02
in_deals_pc                  7.78e-02    0.00e+00
> cat(round(cor(finance_analysis$burt_constraint, finance_analysis$se2_c, use="complete.obs"),3), "\n")
0.457
> vc <- finance_analysis$sa4 %in% se2_avg$sa4
> cat(round(cor(finance_analysis$burt_constraint[vc], finance_analysis$se2_c[vc], use="complete.obs"),3), "\n")
-0.327
> library(readxl)
> f_controls_xlsx <- "C:/Users/s222384959/Downloads/CONTROLS.xlsx"
> religion_raw <- read_excel(f_controls_xlsx, sheet = "relegious affiliation") %>% filter(TIME_PERIOD == 2016) %>%
+   mutate(category = str_trim(str_extract(`Data Item`, "(?<=:).*(?= \\()")), sa4 = as.character(ASGS_2021))
> religion_wide <- religion_raw %>% filter(!str_detect(category, "Inadequately")) %>%
+   dplyr::select(sa4, category, OBS_VALUE) %>% pivot_wider(names_from = category, values_from = OBS_VALUE) %>%
+   mutate(across(-sa4, ~ replace_na(.x, 0)))
> n_cats <- ncol(religion_wide) - 1
> religious_diversity <- religion_wide %>% rowwise() %>%
+   mutate(total = sum(c_across(-sa4), na.rm = TRUE),
+          entropy_raw = { props <- c_across(-sa4) / total; props <- props[props > 0]; -sum(props * log(props)) },
+          religious_diversity = (entropy_raw / log(n_cats)) * 100) %>% ungroup() %>% dplyr::select(sa4, religious_diversity)
> chk <- desc_vars_homeo %>% left_join(religious_diversity, by = "sa4")
> cat("N:", nrow(chk), "| r(religious diversity, ethnic diversity) =", round(cor(chk$religious_diversity, chk$entropy_c, use = "complete.obs"), 3), "\n")
N: 80 | r(religious diversity, ethnic diversity) = 0.943
> library(readxl)
> f_controls_xlsx <- "C:/Users/s222384959/Downloads/CONTROLS.xlsx"
> religion_raw <- read_excel(f_controls_xlsx, sheet = "relegious affiliation") %>% filter(TIME_PERIOD == 2016) %>%
+   mutate(category = str_trim(str_extract(`Data Item`, "(?<=:).*(?= \\()")), sa4 = as.character(ASGS_2021))
> religion_wide <- religion_raw %>% filter(!str_detect(category, "Inadequately")) %>%
+   dplyr::select(sa4, category, OBS_VALUE) %>% pivot_wider(names_from = category, values_from = OBS_VALUE) %>%
+   mutate(across(-sa4, ~ replace_na(.x, 0)))
> n_cats <- ncol(religion_wide) - 1
> religious_diversity <- religion_wide %>% rowwise() %>%
+   mutate(total = sum(c_across(-sa4), na.rm = TRUE),
+          entropy_raw = { props <- c_across(-sa4) / total; props <- props[props > 0]; -sum(props * log(props)) },
+          religious_diversity = (entropy_raw / log(n_cats)) * 100) %>% ungroup() %>% dplyr::select(sa4, religious_diversity)
> chk <- desc_vars_homeo %>% left_join(religious_diversity, by = "sa4")
> cat("N:", nrow(chk), "| r(religious diversity, ethnic diversity) =", round(cor(chk$religious_diversity, chk$entropy_c, use = "complete.obs"), 3), "\n")
N: 80 | r(religious diversity, ethnic diversity) = 0.943
> raw_vars <- c("homeostasis","new_firms_percapita","se1_avg","se2_avg",
+               "creative_class_share","entropy","density_100",
+               "log_med_income_avg","unemployment_rate_avg","age_25_64_avg",
+               "employment_rate_avg","specialisation_index",
+               "burt_constraint","in_deals_pc")
> d80 <- desc_vars_homeo %>%
+   select(sa4, homeostasis, se1_avg, creative_class_share, entropy, density_100,
+          log_med_income_avg, unemployment_rate_avg, age_25_64_avg,
+          employment_rate_avg, specialisation_index) %>%
+   inner_join(desc_vars_fin %>% select(sa4, new_firms_percapita, se2_avg,
+                                       burt_constraint, in_deals_pc), by = "sa4")
> print(round(t(sapply(d80[raw_vars], function(x)
+   c(N=sum(!is.na(x)), Mean=mean(x), SD=sd(x), Min=min(x), Max=max(x)))), 3))
                      N   Mean     SD    Min    Max
homeostasis          80  0.134  2.117 -3.477  6.442
new_firms_percapita  80  0.028  0.080  0.000  0.622
se1_avg              80  0.072  0.030  0.026  0.150
se2_avg              80  0.279  0.392  0.000  1.000
creative_class_share 80  0.070  0.034  0.020  0.170
entropy              80 57.431 14.908 38.427 90.432
density_100          80  7.475 12.997  0.001 52.725
log_med_income_avg   80  6.763  0.209  6.397  7.296
unemployment_rate_avg 80  5.702  1.244  3.173  9.188
age_25_64_avg        80 52.234  3.221 46.350 65.213
employment_rate_avg  80  0.475  0.068  0.142  0.600
specialisation_index 80 12.591  4.744  5.638 29.162
burt_constraint      80  0.430  0.401  0.000  1.000
in_deals_pc          80  0.010  0.016  0.000  0.093
>
```